SCORE/XX:

Math 304
Linear Algebra
Wednesday, Oct. 25, 2017

NAME:

PRACTICE MIDTERM II

Please show your work and write only in **pen**. Calculators and notes are forbidden. Only 1 brain (your own) allowed per exam. Please also note that there are **X** problems, some of which have multiple parts.

1: Please find the characteristic polynomial of $\begin{bmatrix} 1 & 2 & 3 \\ -1 & 2 & -4 \\ 2 & 3 & 5 \end{bmatrix}$.

2: Please write the solution to the linear system $\begin{bmatrix} 1 & 0 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ as a vector of linear functions in 2 parameters. Then describe how you can find 2 vectors in $\mathbb{R}^{5\times 1}$ whose linear combinations describe all solutions of the preceding linear system.

3: Please write the solution to the linear system $\begin{bmatrix} 1 & 0 & 0 & 0 & 7 \\ 0 & 0 & 0 & 1 & -13 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ as a vector of linear functions in 3 parameters. Then describe how you can find 3 vectors in $\mathbb{R}^{5\times 1}$ whose linear combinations describe all solutions of the preceding linear system.

4: Please write the solution to the linear system $\begin{bmatrix} 1 & -12 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ as a vector of linear functions in 3 parameters. Then describe how you can find 3 vectors in $\mathbb{R}^{5\times 1}$ whose linear combinations describe all solutions of the preceding linear system.

5: Assume $\theta \in \mathbb{R}$. Please find the eigenvalues (and corresponding eigenvectors) for $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$, as a function of θ.

6: (a) Please find a 2×2 matrix, that is *not* the zero matrix, whose only eigenvalue is 0.
(b) Please prove that $\begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$ is *not* diagonalizable, i.e., there is *no* invertible 2×2 matrix S with $S\begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}S^{-1}$ diagonal. **Hint:** First find the eigenvalues $\{\lambda_1, \lambda_2\}$ of $\begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$. Then prove that the identity $S\begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} S$ is impossible to attain.

7: Please express $\begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}^{128}$ in the form $\begin{bmatrix} a\alpha^{128} + b\beta^{128} & c\alpha^{128} + d\beta^{128} \\ r\alpha^{128} + s\beta^{128} & u\alpha^{128} + v\beta^{128} \end{bmatrix}$ (for some explicit constants $a, b, c, d, r, s, u, v, \alpha, \beta$).

8: Please express $e^{\begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}}$ in the form $\begin{bmatrix} ae^{\alpha} + be^{\beta} & ce^{\alpha} + de^{\beta} \\ re^{\alpha} + se^{\beta} & ue^{\alpha} + ve^{\beta} \end{bmatrix}$ (for some explicit constants $a, b, c, d, r, s, u, v, \alpha, \beta$).

9: Suppose $A \in \mathbb{R}^{2\times 2}$ has trace 11 and determinant 18. What are the eigenvalues of A?

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10: Suppose $A \in \mathbb{R}^{n\times n}$ is invertible and has a nonzero eigenvalue λ. Please prove that A^{-1} has $\frac{1}{\lambda}$ as an eigenvalue.

11: (a) Please prove that if $A \in \mathbb{C}^{n\times n}$ is a matrix satisfying $A^2 = A$ then any eigenvalue of A must be 0 or 1.

(b) Please prove that if $A \in \mathbb{C}^{n\times n}$ satisfies $A^k = \mathbf{O}$ (the zero matrix) for some integer k then all the eigenvalues of A are 0.

(c) Please find the first 4 powers of $\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$.

12: Suppose $A \in \mathbb{C}^{5\times 5}$, I is the 5×5 identity matrix, and A has eigenvalues 4, 2, 7, 9, and -1. Please find the eigenvalues of $A - \sqrt{2}I$ and prove why your answer is correct.

13: Suppose $M \in \mathbb{R}^{3\times 3}$ is a matrix satisfying $M \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}$ and $M \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$.

(a) Please find two distinct eigenvalues of M. (b) Please find two distinct eigenvectors of M.